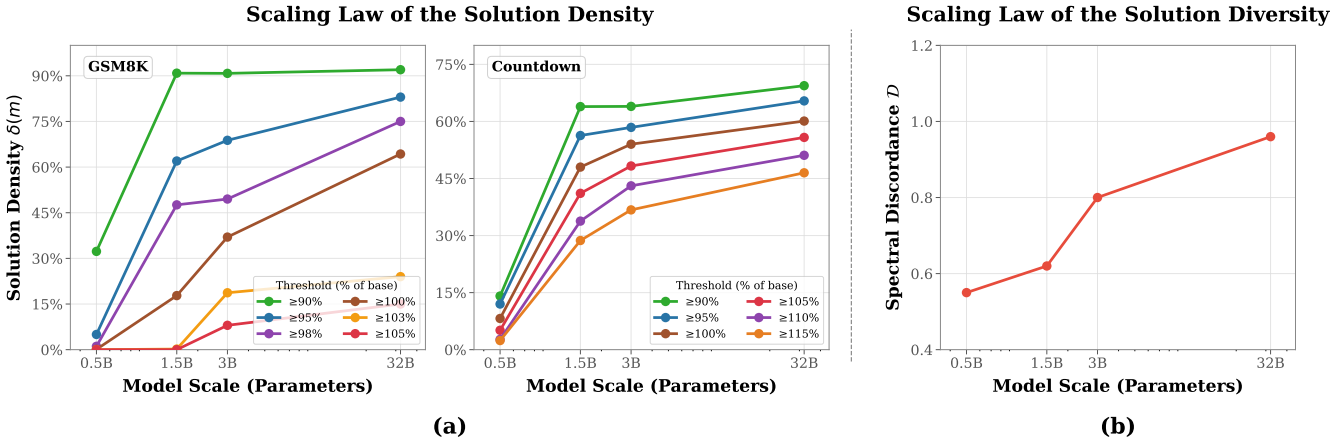

Figure 3: Scaling laws of solution density and diversity (using *Qwen-2.5* instruction tuned models). (a) Solution density increases with model scale, showing that larger models have a higher fraction of good solutions. (b) Spectral discordance across model scales, measuring solution diversity. Together, these results demonstrate that larger models have both denser and more diverse solution landscapes in the neighborhood around their pretrained weights.

indicate that, for large-scale models, the pretrained weights reside within a dense basin populated by abundant high-quality solutions, whose density scales with model size.

2.2 Solution Diversity: Are the Sampled Perturbations Specialists or Generalists?

Do all perturbations help (or hurt) in the same way? We compare two hypotheses:

1. Hypothesis 1 (generalists): The pretrained weights are in fact a poor model for the suite of downstream tasks we are testing; there is an all-around better model in the vicinity of these weights.
2. Hypothesis 2 (specialists): The pretrained weights are a "jack of all trades, master of none"; perturbations can improve a given task because they are specialists for that task, improving its performance while hurting performance on other tasks.

We test these hypotheses using the following measure of specialization:

Definition 2.2 (Spectral Discordance)**.** Let $\mathbf{P} \in [0,1]^{N \times M}$ denote the percentile-rank matrix over N seeds and M tasks, and let $\mathbf{C} \in \mathbb{R}^{M \times M}$ be the Pearson correlation matrix of its columns. We define the *Spectral Discordance* as:

$$\mathcal{D} = 1 - \frac{1}{M(M-1)} \sum_{j \neq k} \mathbf{C}_{jk} \tag{2}$$

where $\mathbf{C}_{jk}$ denotes the correlation between tasks j and k. A value of $\mathcal{D} \to 1$ implies orthogonal task rankings (specialists), while $\mathcal{D} \to 0$ implies parallel rankings (generalists).

Remark. Theoretically, $\mathcal{D}$ is bounded in the interval $[0, \frac{M}{M-1}]$. The upper bound corresponds to the limit of anti-correlation. We provide the derivation in Appendix H.

We evaluate 500 random weight perturbations across seven tasks in four domains: mathematical reasoning (Countdown, GSM8K, MATH-500, OlympiadBench), code generation (MBPP), creative writing (ROCStories), and chemistry (USPTO). For each perturbation, we compute its percentile rank relative to the population. We then quantify diversity using the *Spectral Discordance* $\mathcal{D}$ over all the perturbations. Figure 3(b) shows the result over models of different size in the Qwen2.5 family: $\mathcal{D}$ increases monotonically with model size, indicating that the solutions surrounding larger models become increasingly disjoint in their capabilities. This supports Hypothesis 2: the sampled solutions are specialists.

To visually unpack the structure of this discordance, Figures 4 and 7 present further analyses:

(1) Performance Spectra (Figure 4 Left): We plot the percentile ranks of individual perturbations across tasks. The resulting lines are "spiky" rather than flat. This visually demonstrates specialization and diversity among the sampled perturbations.

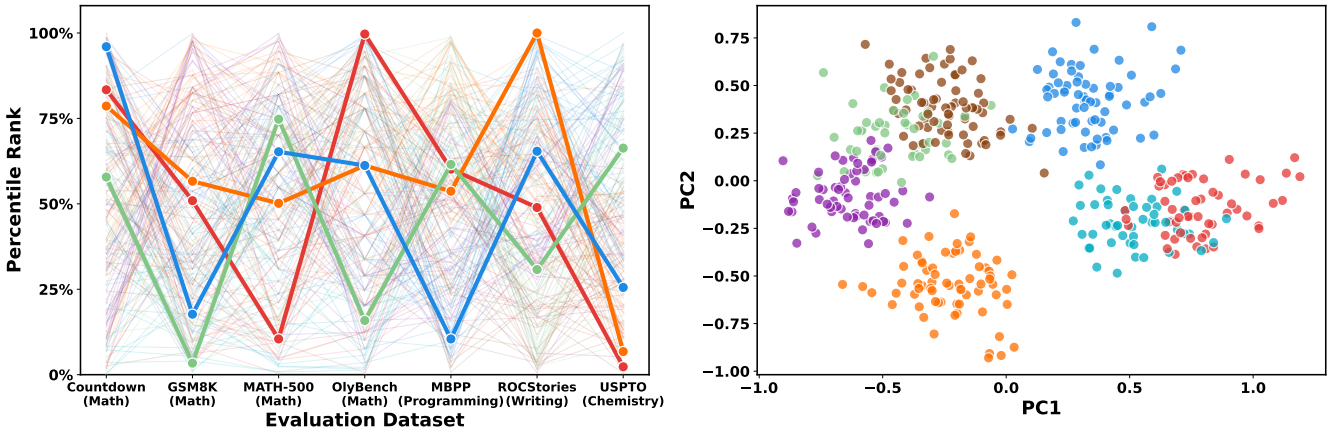

Figure 4: Performance spectra and clustering of random seeds. Sampled vectors possess diverse areas of expertise, with individual seeds specializing in specific tasks. (Left) Performance of 100 random seeds across seven evaluation datasets. Each line represents a specific seed, with four lines highlighted as examples. (Right) PCA visualization of these performance vectors, where seeds with similar behavior cluster together into different groups.

(2) PCA Projection (Figure 4 Right): We project the 7-dimensional performance vectors into 2D and apply K-means clustering. The emergence of distinct clusters confirms that there are multiple kinds of experts within the distribution. Perturbations within a cluster share specific strengths (e.g., excelling at math but failing at chemistry), while perturbations in different clusters offer complementary capabilities.

(3) The level of diversity is also visible in Figure 7. Under the column labeled "RGB", we plot three different task-accuracy landscapes in the R, G, and B channels respectively. The fact that there is a mottled, multi-colored appearance indicates that the landscapes are largely uncorrelated (we would expect to see shades of gray if all tasks behaved the same).

These findings reveal that the local weight space is populated by diverse specialists. This raises a key question: can we exploit this landscape by simply sampling these complementary experts and aggregating their strengths? We will return to this question in Section 4.

3 A Minimal Setting In Which Thickets Emerge: Autoregressive Modeling of 1D Signals

What leads to thickets emerging? To elucidate the cause, we replicate the phenomenon in a minimal setting.

We define a training distribution that is a mixture of several random function types (sinusoidal, linear, harmonic, sigmoidal, and sawtooth and square waves), each of which maps $\mathbb{R} \to \mathbb{R}$ and takes on random settings for the function parameters (e.g., phase and amplitude for sinusoids, slope and intercept for lines). We pretrain a next-value predictor on these functions, using a multilayer perceptron $f_\theta : \mathbf{y}_{\text{CTX}} \to y_{\text{NEXT}}$, where $\mathbf{y}_{\text{CTX}}$ is a preceding context window and y_{NEXT} is the next value of the target function. This model can generate predictions by autoregressive rollout given an initial observed context. We probe this model with a simple linear test signal. Is this signal well-modeled by sampled perturbations near the pretrained weights?

We sample $N = 1000$ random Gaussian parameter perturbations from $\epsilon \sim \mathcal{N}(0, \sigma = 0.002)$. In Figure 5, we show autoregressive rollouts of the base and perturbed models given the test context. The blue line is the test function, with solid blue as the observed context and dashed blue as the ground truth continuation. The gray lines are predictions given by different random perturbations. The base, unperturbed model's predictions are shown in black. The top 5 perturbations that best fit the blue line are shown in red.

We compare three pretraining settings: 1) no pretraining (Xavier initialization, Glorot & Bengio (2010)), 2) pretraining on all signal types, 3) pretraining on linear signals. These three settings lead to three different regimes: 1) the needle in the haystack regime, where small perturbations of the weights have negligible effect on the functional shape; good solutions are far away from this initialization, 2) the thicket regime, where different

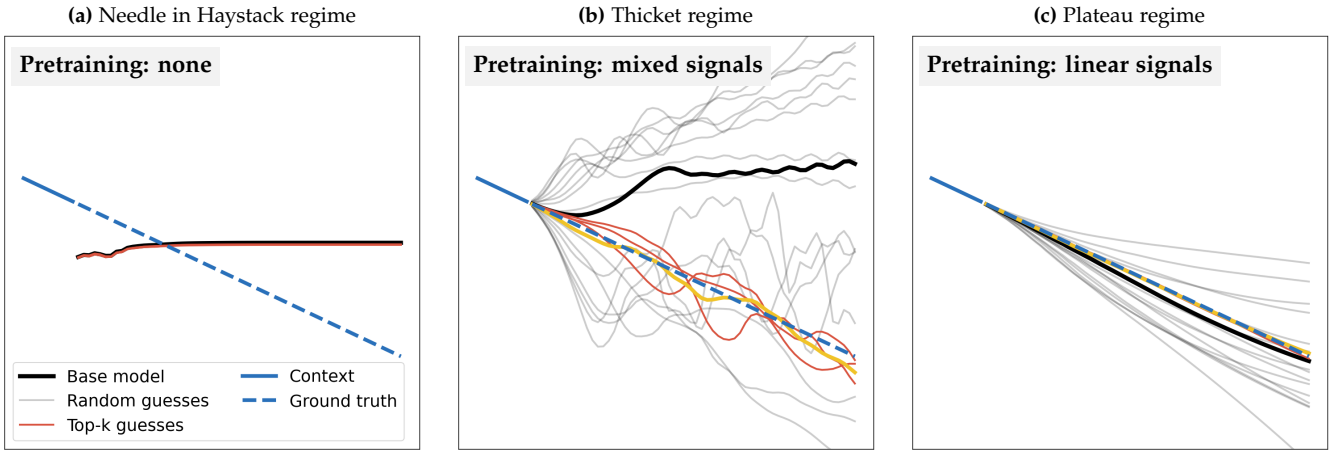

Figure 5: Pretraining a model of 1D signals, then probing the local neighborhood around the pretrained weights by random guessing $N = 1000$ Gaussian perturbations. The plot shows the autoregressive predictions of a particular linear function (dashed blue line), given an observed context (solid blue line). Gray lines: random f_θ's; Red lines: top-K f_θ's. The figure shows three regimes: (a) No pretraining leads to needle-in-the-haystack search, (b) pretraining on several signal types leads to a thicket, (c) pretraining on just linear functions achieves nearly perfect predictions at pretraining time, hence post-training is at ceiling.

perturbations search over many possible continuations following the kinds of functions seen during pretraining, and 3) a "plateau" regime, where the pretrained weights are already a minimizer of the test task, and random guessing can provide no further benefit.

This experiment demonstrates that the phenomena we are observing are not exclusive to LLMs, and show up in simpler models as well. What appears to be critical is that the base model is pretrained on a variety of signal shapes. Too little pretraining results in no nearby solutions and pretraining on just one signal type results in nearby weights showing very little functional diversity. See Appendix F for more examples, including generalization to held out test signals, the effect of ensembling, and settings of weight initialization and pretraining type.

4 A Practical Algorithm: Random Guessing & Ensembling (RandOpt)

The fact that valid solutions are easy to find but possess non-overlapping strengths, suggests that rather than searching for a single global minimum, we might instead sample broadly and aggregate the predictions. We therefore explore an algorithm, which we call RandOpt, that *randomly guesses* a set of N weight perturbations, and then *ensembles* the top K.

Let $f_{\boldsymbol{\theta}} : \mathcal{X} \to \mathcal{Y}$ denote a base model parameterized by weights $\boldsymbol{\theta} \in \mathbb{R}^d$. We introduce perturbations via a noise vector $\boldsymbol{\epsilon}$ sampled from a standard Gaussian distribution, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. The magnitude of each perturbation is controlled by a noise scale $\sigma \in \mathbb{R}^+$. To explore neighborhoods at different scales, we use a *set of scaling factors* $\Sigma = \{\sigma_1, \ldots, \sigma_M\}$. A perturbed model instance $\boldsymbol{\theta}'$ is determined by a random seed s and a selected scale $\sigma \in \Sigma$:

$$\boldsymbol{\theta}' = \boldsymbol{\theta} + \sigma \cdot \boldsymbol{\epsilon}(s) \tag{3}$$

where $\boldsymbol{\epsilon}(s)$ denotes the noise vector generated by seed s.

RandOpt operates in two phases: *Random Guessing* (Training) and *Ensembling* (Inference), which are given in pseudocode in Algorithm 1 and described in math below:

Training: Random Guessing and Checking. We sample a population of N random seeds $\{s_1, \ldots, s_N\}$, and a corresponding noise scale per seed, $\{\sigma_1, \ldots, \sigma_N\}$, with all σ_i sampled uniformly from Σ. These give a collection of N parameter vectors $\theta_i = \theta + \sigma_i \boldsymbol{\epsilon}(s_i)$. Each corresponding model $f_{\boldsymbol{\theta}_i}$ is evaluated on a small training or validation set $\mathcal{D}_{\text{train}}$ to obtain a performance score v_i. We then select the top-K performing models based on these scores:

$$\mathcal{I}_{\text{top}} = \underset{i \in [N]}{\arg\,\text{topK}}(v_i) \tag{4}$$